

IMPLEMENTATION OF A SLOSH MECHANICAL MODEL DURING SPACECRAFT APPROACH AND DOCKING

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This paper presents the application of a slosh mechanical model to assess the effects of propellant slosh on the success of a docking maneuver between two spacecraft. The slosh model comprises a free-moving particle within a constraint surface with penalty method-based contact dynamics and an inert particle that augments the mass properties of the spacecraft. The parameters of the slosh model are tuned with computational fluid dynamics (CFD) simulations of a representative scenario. The slosh model is then integrated into the approach and docking phase simulations. Results indicate that the slosh model can be used to inform when and how propellant slosh affects the performance and robustness of the system during the approach and docking phases.

INTRODUCTION

The topic of this paper is the rendezvous, proximity operations, and docking between two orbital spacecraft. Specifically, the focus is the approach and docking between an active chaser spacecraft containing liquid propellant and a passive target spacecraft. During docking, the docking system must first arrest the relative motion between the spacecraft, align the docking ports, and then structurally mate the vehicles together. The liquid propellant on board the chaser spacecraft may slosh, resulting in relative vehicle motions that damage the docking mechanism during operation and cause mission failure. Therefore, an analysis of the effects of propellant slosh on docking is prudent.

In this work, the approach and docking phases are handled by two separate simulations. A flight dynamics simulation in MATLAB®[¶] and Simulink® was used to model the approach phase and includes six-degree-of-freedom (DOF) rigid-body dynamics, orbital mechanics, and the guidance and control of the chaser spacecraft. A high-fidelity multi-body dynamics simulation in Adams^{||} was used to model the detailed docking phase between the chaser and target spacecraft. Due to its complexity and ensuing runtime, it was deemed intractable to perform a comprehensive Monte Carlo analysis using the Adams model. As a result, the approach of the chaser vehicle was simulated with

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the 6-DOF spacecraft simulation and the worst cases — identified by a combination of misalignments and rates at the time of initial contact — were used to down-select the initial conditions for the docking simulation. A mechanical slosh model that comprises a single inert particle and a single translating particle was used to simulate the motion of the propellant and its interaction with the chaser during both the approach and docking phases; the original version of this slosh model was first implemented under NASA’s Apollo program.^{1–3} This paper details the development, tuning, and integration of the slosh model into the approach and docking phase simulations.

The contributions of this paper are: 1) an improvement to the existing single-particle mechanical slosh model for low and micro-g; 2) the tuning of this slosh model to CFD simulation data in a micro-g environment; and 3) the application of the model to estimate and assess the effects of large amplitude propellant slosh during approach and docking between two spacecraft.

The remainder of the paper includes the following sections: the Background section discusses the state of the art; the Slosh Model section outlines the improved particle slosh model; the Model Parameters section describes how the parameters of the slosh model were tuned; the Simulation and Results section presents the results of the approach and docking phase simulations; and, finally, the Conclusion section presents some closing remarks.

BACKGROUND

During approach and docking operations in orbit, the chaser spacecraft is in a micro-g environment and subject to thruster forces from the control system and reaction forces from the docking mechanism. As a result, the propellant may undergo large amplitude sloshing and liquid reorientation. For the highest level of fidelity, co-simulation between computational fluid dynamics (CFD) simulations and flight dynamics is standard practice. For this, the dynamics of the closed-loop vehicle are simulated and, at each time step, vehicle states are passed to the CFD simulation to simulate the propellant motion. Any interactions between the propellant and the tank walls are passed onto the flight dynamics simulation as forces and torques on the vehicle. The iterative nature of this approach with CFD in the loop is computationally expensive and impractical for performing a Monte Carlo analysis involving many simulations. As such, low-order mechanical models are preferable for this purpose because they are faster than CFD. These models are low order and numerically efficient compared to CFD simulations. The goal is to maintain sufficient model fidelity such that the forces and motions of the liquid can be predicted in the moments following known initial conditions. The trade off is that mechanical models are unable to simulate complex fluid motions, and accuracy is expected to decrease with time more sharply than CFD simulations.

Simple harmonic oscillators, including mass-spring-damper systems or pendulums, are typically used for high-g hydrodynamic regimes ($Bo > 100$), and may be extended to low-g environments ($\sim 30 < Bo < 100$).⁴ These models can capture both linear and nonlinear motions depending on the pedigree of the model used.⁵ The accuracy of these internal oscillators, however, begin to break down as the gravitational acceleration decreases. They are therefore not expected to effectively predict the potential slosh motions of the present docking application which is characterized by large amplitude sloshing and liquid reorientation (i.e., $We \gg 1$) in a micro-g environment. Alternative mechanical models to the internal oscillators have been developed to simulate large amplitude sloshing and liquid reorientation in low or micro-g ($Bo < \sim 30$) environments.^{1,6–10} The model used in this analysis is selected for its simplicity and heritage.^{1,6–8}

Berry and Tegart developed a mechanical analog which represents the liquid as a point mass

moving on an ellipsoidal constraint surface.⁶ Test data from drop tower tests were used to provide limited model validation, and the correlation between test and analytical results indicates that the mechanical analog is capable of predicting overall force trends measured during the short duration testing. While the trends were similar, the magnitude of the slosh forces did not compare well between model and test. Berry and Tegtart note that future tuning of the model—as would be conducted by Good et al., Zhou and Huang, and as in this work—may allow for better correlation between model and test data. Good et al. deployed the same constrained particle model presented in Berry and Tegtart to analyze two slosh-related problems.⁷ The first regards analyzing the effects of high-g slosh on the stability of a spacecraft with a unique propellant tank orientation during trajectory correction maneuvers. The second regards analyzing the effects of slosh on the despin of the spacecraft and the possibility of recontact with the launch vehicle upper stage. The despin and recontact analysis involved a novel configuration of the constrained particle model whereby multiple particles are used to simulate the propellant motions. The models were tuned using CFD simulation and experimental drop-tower test results. Zhou and Huang presented a framework that allowed the particle to come free from the constraint surface and model subsequent contacts via penalty method-based contact dynamics.⁸ This model was tuned against a CFD simulation and subsequently compared to the on-orbit experimental results from the Sloshsat FLEVO satellite.^{11,12}

The environment examined by Zhou and Huang takes place in a relatively quiescent low-g environment with steady sinusoidal accelerations and body rates. During the docking maneuver, however, the gravitational environment will be in the micro-g hydrodynamic regime with aperiodic disturbances from the control thrusters during the approach phase and from the docking mechanism during the docking phase. Therefore, tuning the model presented by Zhou and Huang with data from a representative acceleration environment would improve its applicability to the present docking problem.

SLOSH MODEL

This section describes the development and tuning of the mechanical slosh model. The model implemented here, shown in Figure 1, is based on the works of Berry and Tegtart, Merchant et al.,¹ Good et al., and Zhou and Huang. The slosh model approximates the motion of the liquid within a tank as two lumped mass particles. One of the particles is fixed at the center of the tank and augments the spacecraft mass properties; this particle captures the effective inertia added to the spacecraft by the inert portion of liquid. The other particle can freely translate within a boundary called the constraint surface and is intended to capture the forces imparted on the vehicle by the sloshing motions of the liquid. When the moving particle contacts the constraint surface, a restoring force is computed via penalty method-based contact dynamics to return the particle to the surface. Notable differences from the model presented by Zhou and Huang are 1) the introduction of the inert particle to augment the spacecraft mass properties; 2) the implemented viscous friction model; and 3) the smoothed contact dynamics.

Contact Dynamics

In Figure 2, the slosh particle outside is the constraint surface. When the particle exits the constraint surface, a restoring force F_n may be estimated with the expression

$$F_n = -kx - c(x, \dot{x})\dot{x}, \quad (1)$$

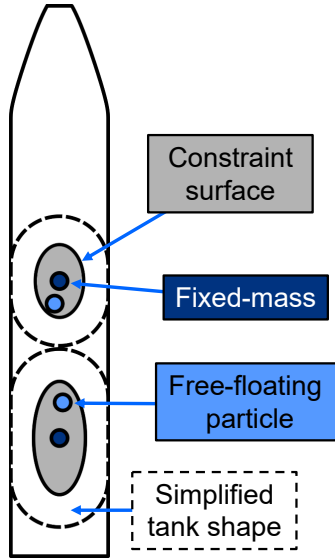


Figure 1: The slosh mechanical model.

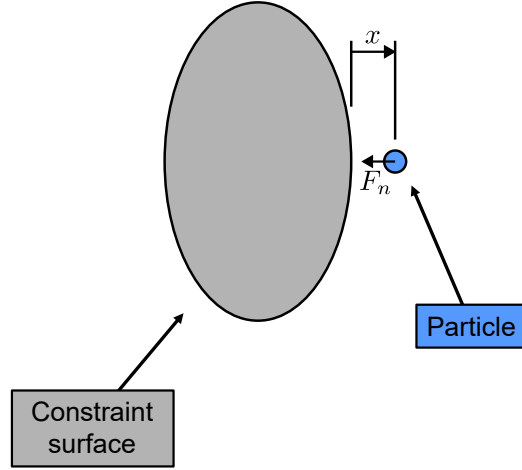


Figure 2: Computing the restoring force.

where F_n is the magnitude of the normal contact force, x is the penetration depth (perpendicular to the constraint surface; positive radially outwards), $\dot{x}$ is the rate of the particle that is perpendicular to the constraint surface, k is the stiffness coefficient, and $c(x, \dot{x})$ is the smoothed damping coefficient. Without the smoothed damping coefficient, which is discussed below, the contact response could be described by a transfer function of the form

$$G(s) = \frac{1}{ms^2 + 2m\zeta\omega_n s + m\omega_n^2}, \quad (2)$$

where m is the mass of the particle, ω_n is the natural frequency, and ζ is the damping coefficient. A critically damped system ($\zeta = 1$) is assumed since this is analogous to a perfectly inelastic collision spread out over time. This leads to the relationship

$$k = m\omega_n^2, \quad c_{crit} = 2m\omega_n, \quad (3)$$

where c_{crit} is the coefficient for the critically damped system. The implications of a critically damped system are that the kinetic energy within the particle model that is associated with initial contact is dissipated, and therefore the particle leaves the surface only due to spacecraft rigid body motion. Disturbances such as reaction control system jets, docking forces, or slosh forces from another tank would be expected to induce particle separation or additional loads.

The mass of the moving particle is a function of the total mass of the liquid. The mass of the particle is expressed as

$$m = EMF \times m_{prop}, \quad (4)$$

where m_{prop} is the total mass of the propellant in the tank and $EMF \in (0, 1]$ is the effective mass fraction. EMF is a tuning parameter which represents the portion of the liquid that induces the sloshing forces and moments on the spacecraft. The remaining fraction of liquid (i.e., $1 - EMF$) makes up the non-sloshing portion of the liquid, which the inert particle represents.

Without smoothing, there will be a discontinuous jump in contact force when the moving particle transitions from freely drifting to in-contact. This is because $F_n = 0$ at the instant prior to contact and $F_n = -c_{crit}\dot{x}$ at the instant just after contact. This can lead to simulation numerical sensitivity to time step size. Therefore, a smoothing function is applied to the damping coefficient to alleviate the discontinuous jumps in contact force.

The smoothed damping coefficient implemented here is a function of penetration depth x and penetration rate $\dot{x}$. This function is expressed as

$$c(x, \dot{x}) = smoothstep(\dot{x}, 0, c_{crit}, \dot{x}_d, smoothstep(x, 0, 0, x_d, c_{crit})), \quad (5)$$

where x_d and $\dot{x}_d$ are the smoothing ranges for the penetration depth and rate, respectively, and

$$v = smoothstep(u, u_L, v_L, u_R, v_R) \quad (6)$$

is the clamping and interpolation function shown in Figure 3, where u and v are the input and output, respectively, and u_L, v_L, u_R, v_R are the function parameters. The clamping and interpolation function is expressed as

$$v = \begin{cases} v_L, & u \leq u_L \\ v_L + (v_R - v_L)S_1, & u_L < u < u_R \\ v_R, & u \geq u_R \end{cases} \quad (7)$$

where S_1 is the cubic Hermite polynomial

$$S_1(\delta) = 3\delta^2 - 2\delta^3, \quad \delta = \frac{u - u_L}{u_R - u_L}. \quad (8)$$

The *smoothstep* function is used extensively in simulations such as Adams. Note the *smoothstep* function will modify the response of the contact dynamics such that it does not match the response of a critically damped system exactly.

The form of Equation (5) is designed so that the damping coefficient smoothly ramps up to c_{crit} as the particle penetrates into the constraint surface but also so the coefficient does not ramp down to 0 as the particle makes its way out. In other words, we want $c(x = 0, \dot{x} > 0) = 0$ but also $c(x = 0, \dot{x}_d < 0) = c_{crit}$. This is not possible with only a single execution of *smoothstep*. Instead, we can accomplish this by executing two consecutive *smoothstep* functions, as shown in Equation (5), where the inner *smoothstep* smoothes the damping coefficient as a function of x and the outer smoothes as a function of $\dot{x}$.

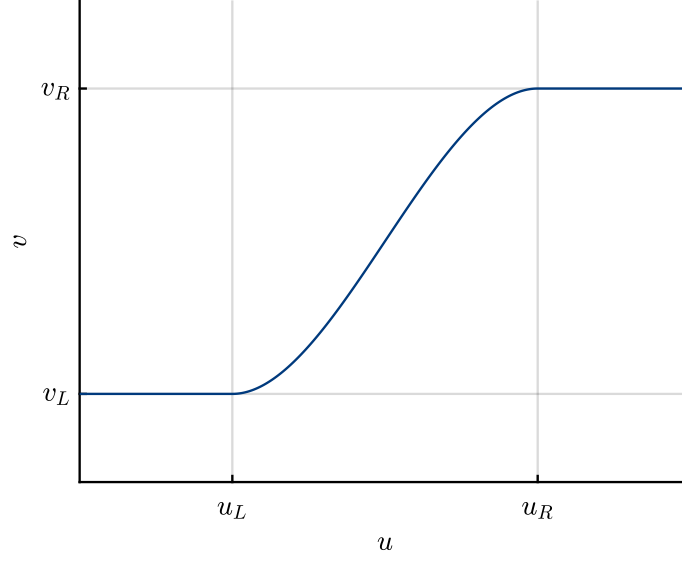


Figure 3: Clamping and interpolation function.

Energy Dissipation

Energy dissipation is an important characteristic for modeling slosh. Energy is dissipated when the liquid contacts and moves along the surface of the walls. The energy dissipation of the liquid moving perpendicular to the wall is captured by the damping in the contact dynamics. This section describes the viscous friction force, F_t , as the particle translates tangent to the constraint surface. In general, it is difficult to correctly capture the viscous dissipative force. An in-depth investigation into the form of the viscous dissipative force was beyond the scope of this work.

Berry and Tegart comment about the viscous dissipative force stating that “its characteristics are not well-defined, but previous investigations¹³ indicate that significant parameters may be kinematic viscosity, characteristic length, and gravitational environment.” Berry and Tegart represent the magnitude of the viscous friction force as a linear combination of the normal force and tangential velocity expressed as

$$|F_t| = \mu F_n + \eta v_{tan}, \quad \mu, \nu \geq 0, \quad (9)$$

where F_n is the normal force, v_{tan} is the velocity tangent to the constraint surface, and μ and η are parameters input to the model to tune the response. Ultimately, Berry and Tegart set $\mu = 0$ and select several values for η to match the forcing from the model to the available data from the drop tower test. As noted previously, the match between the results from the mechanical model and the test data are limited to general trends in the time histories despite tuning the model parameters to improve the comparison. Berry and Tegart summarize as follows: “A study of the comparisons indicates that the assumed form for the viscous dissipative force [Equation (9)] is not optimal for the simulation. As mentioned in Chapter III, the form of this force is not well known. An in-depth investigation into its form was beyond the scope of this study, however, indications are that it may be nonlinear.”

Good et al. and Zhou and Huang use the same model for energy dissipation, but do not provide insight into the values of η used in the model. Good et al. and Zhou and Huang made no connections

between the fluid properties used in the CFD simulation and the selection of the model parameter, η .

A viscous dissipative force used here was introduced in Merchant et al. to simulate energy dissipation in the Apollo-era service module after jettison to assess the possibility of recontact. This model is expressed as

$$F_t = -C_f F_n v_{tan}, \quad (10)$$

where C_f is the coefficient of viscous friction, F_n is the normal force as described in the contact dynamics, and v_{tan} is the magnitude of the velocity of the particle tangent to the constraint surface. This model is similar to Equation (9); however, F_t is proportional to F_n and v_{tan} instead of being a linear combination of the two. A value of $C_f = 0.01 \frac{1}{ft/s}$ ($0.03 \frac{1}{m/s}$) was used by Merchant et al. and “obtained from the classical solution of the energy losses for turbulent fluid flow in curved pipes.”

MODEL PARAMETERS

The previous section describes the slosh model including contact forcing and energy dissipation that is used for the assessment. This section describes how the model parameters are tuned using CFD simulation results. To summarize, the model parameters are:

- ω_n , the natural frequency of the contact dynamics
- EMF , the effective mass fraction of the liquid
- x_d , the smoothing depth of the contact dynamics
- $\dot{x}_d$, the smoothing rate of the contact dynamics
- C_f , the coefficient of viscous friction

The goal is to have a particle model that provides slosh force outputs that have the correct order of magnitude and bound the CFD simulations.

The values of x_d and $\dot{x}_d$ are generally selected to be greater than the expected penetration depth and rate, respectively. Therefore, for the docking analysis, $x_d = 1$ m is the same order of magnitude as the tank diameter of the tanks, and $\dot{x}_d = 0.1$ m/s since this is a notional upper bound on the relative velocity of the two vehicles during the approach phase.

The values of ω_n and EMF are tuned with CFD simulations and are discussed below. The CFD simulations only consider motion along one direction — the longitudinal axis of the tanks. If lateral translational and rotational motions were simulated using CFD, the results would have been applicable for tuning C_f . However, in the absence these motions, several values of C_f were compared during a simple spacecraft slewing maneuver using the 6-DOF flight dynamics simulation. As a result of this testing, it appeared that $C_f = 0.1 \frac{1}{m/s}$ is a reasonable and conservative value.

Computational Fluid Dynamics Simulations

This section discusses the tuning of ω_n and EMF (slosh model parameters) to improve slosh model matching with the CFD results. The CFD simulation results from four approach and docking scenarios are considered here, with one set of simulation results used for tuning and three used for

validation. We consider only four sets of results due to resource limitations; additional sets of results would have potentially allowed for less conservatism in the slosh model parameters.

The CFD analysis was unvalidated so the resulting force and moment predictions should be considered an estimate. Validating CFD simulations with experimental test data is an active area of research.¹⁴ Nevertheless, the CFD-based estimate of the force output is useful for tuning the lower fidelity particle model. Within the CFD simulations, coupling between the rigid-body and propellant dynamics can increase computation time. Therefore, due to resource constraints, the tanks follow a prescribed trajectory where the effects of the propellant forces on the rigid body motion is not modeled.

Each CFD scenario simulated a notional approach and docking maneuver. The approach phase is simulated by slowing down the tank over approximately 120 s, and the docking phase is simulated by applying an impulse over 1.5 s. As mentioned previously, only translation along the longitudinal axis of the tanks is considered. The liquid is initialized in the aft end of the tanks with a flat free surface. The tank is then left to coast at initial velocity until the free surface reaches a quasi-steady state (about 2000 s). The difference between the four scenarios is the timing and rate of slowdown during each of the events. The maneuver deemed the worst case involved applying the simulated docking impulse at the moment of peak slosh force magnitude, which occurs shortly after the approach phase ends, yielding the largest magnitude slosh forces between the four scenarios. Therefore, this case is used to tune ω_n and EMF such that the particle model provides an upper bound on the slosh forces.

The following procedure was followed to tune the values of ω_n and EMF for the slosh models of each tank:

1. Adjust ω_n such that the 2% settling time of the axial force response is approximately the same as the CFD data.
2. Estimate the peak force, F_{max} , from the CFD axial force response data.
3. Select EMF such that the particle model force response is approximately $2F_{max}$. The multiply-by-two was included to provide margin on the bound of the CFD forcing since the CFD model itself was not validated.

Results

The results of the CFD simulations and tuned slosh model outputs are shown in Figure 4. The simulated docking impulse occurs at approximately $t = 2150$ s and lasts 1.5 s. This period is sufficiently short such that the liquid essentially behaves as a rigid body. This behavior is captured by the inert portion of the mass; therefore, this impulse is not considered when tuning the moving particle, and the next largest peak is selected as F_{max} . As intended, the slosh model upper-bounds the peak and captures the time-domain response of the slosh forces due to liquid reorientation during simulated docking.

SIMULATION AND RESULTS

This section outlines the integration of the slosh model into the approach and docking phase simulations as well as the key implications slosh had on each of these phases. A block diagram of the integration of the slosh model with a spacecraft simulation is shown in Figure 5. The slosh model

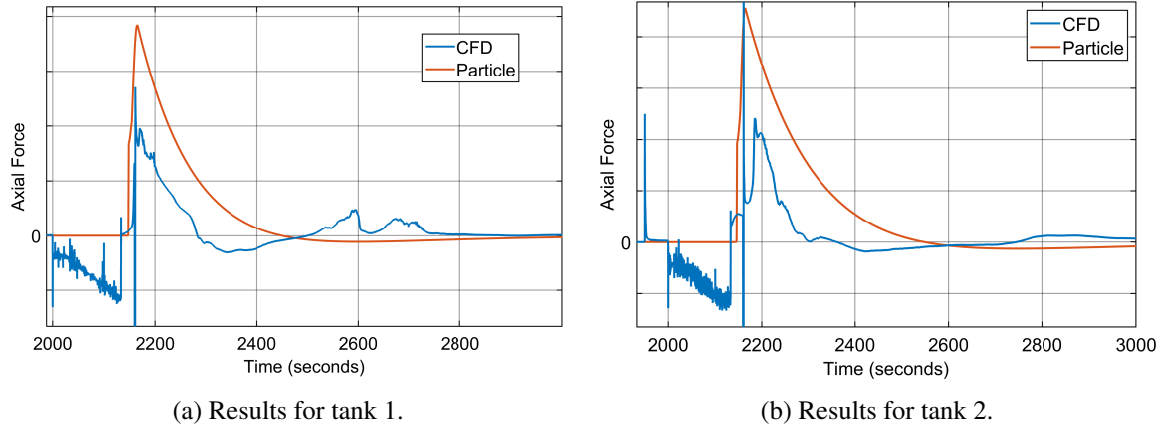


Figure 4: Tuned slosh model outputs to CFD simulation results.

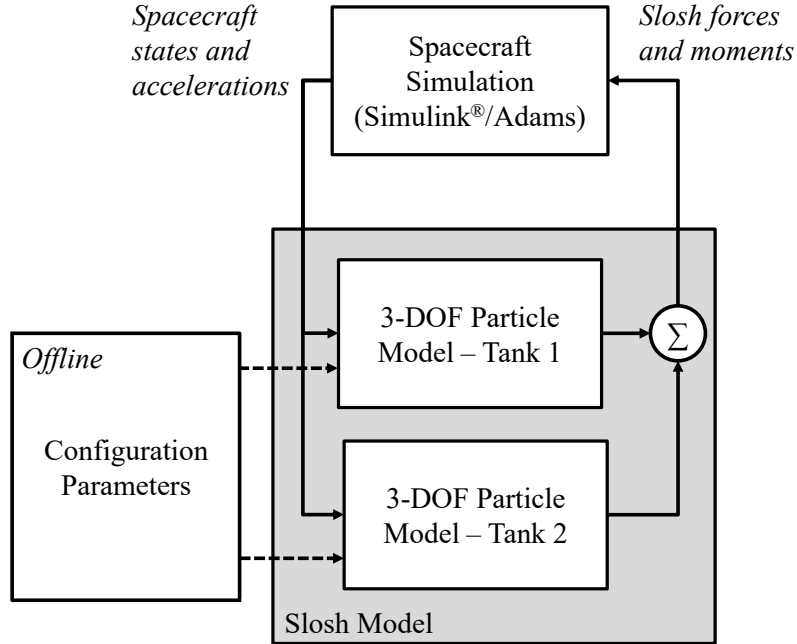


Figure 5: Block diagram of slosh model implementation.

dynamics simulation is fully coupled with the rigid body spacecraft dynamics. Configuration parameters, including the size and location of the constraint surface, mass of the moving particle, and parameters of the contact dynamics, are computed offline and input to the slosh model as constants. Then, at each time step, the slosh model takes as inputs the spacecraft states and accelerations and outputs the forces and moments acting on the spacecraft about the center of mass as a result of slosh motions. Multiple tanks can be simulated (with marginal impact to runtime) whereby the force and moments outputs of each model are summed and applied to the spacecraft.

Approach phase

A 6-DOF simulation, implemented in MATLAB® and Simulink®, is used to analyze the approach phase of the active chaser vehicle to the passive target vehicle. The simulation is of the chaser vehicle and incorporates rigid-body dynamics, orbital mechanics, mechanical slosh particle models for two propellant tanks, representative thruster models, and a flight control system. The chaser vehicle approaches the target vehicle along a prescribed approach trajectory. The feedback control system computes the necessary forces and moments to follow the trajectory while rejecting the disturbances due to gravity-gradient torques and slosh. Two Monte Carlo analyses, each comprising over 2000 runs, were conducted. In the first case, which is the baseline, all propellant mass was modeled as inert and slosh dynamics were ignored. In the second case, the aforementioned slosh model was included. Comparisons of these datasets enable the characterization of slosh motion, identification of trends, and assessment of the impact of slosh on the approach of the chaser to the target.

Each Monte Carlo run initializes the free-floating particles at random, stationary positions within their respective constraint surfaces. A small initial position error of the chaser to the trajectory is included. Three controller gains (lax, nominal, aggressive) are implemented to evaluate how the control system induces sloshing. Controller performance is assessed by comparing the simulated chaser spacecraft state with the desired trajectory over time.

At the conclusion of the approach phase, a snapshot of the chaser spacecraft states is taken for each simulation in the Monte Carlo analysis. The results of the relative lateral misalignment errors between the two spacecraft at the docking interface as a function of controller gains are shown in Figure 6. Figure 6 shows the probability distribution of the lateral misalignment errors for all simulations with slosh enabled and shows the extent of the maximum errors that occur with slosh disabled. The results show that increasing the controller gains resulted in overall better ability of the system to achieve desired misalignment errors at the hand-off between the approach and docking phases. However, the more aggressive controller gains exacerbated the influence of slosh on the extent (in percent change) of the maximum misalignment errors in comparison to the slosh-disabled simulations.

Additionally, the slosh model generated significant forces and torques on the spacecraft at docking in nearly all simulations. The slosh forces were examined following the conclusion of the approach phase when docking would have initiated; in some cases, the slosh forces were sustained for greater than 60 s as the particles reoriented. The magnitudes of the resulting slosh forces aligned with pre-simulation estimates from CFD. Due to the chaotic nature of slosh, single-run simulations proved inadequate for characterizing slosh behavior. Therefore, it is recommended to perform a Monte Carlo analysis — like the one presented here — to effectively assess the influence of slosh dynamics on the performance and robustness of the control system for docking in micro-g.

Docking phase

The relative misalignments and velocities at the docking interface at the end of the approach phase are evaluated against the Initial Contact Condition (ICC) requirement limits as a measure of docking success. The ICCs from the results of the 4000 approach simulations included cases outside of the lateral misalignment limits both with and without slosh effects enabled. However, the maximum lateral misalignments with slosh enabled were 23% to 48% larger than those with slosh disabled, depending on the controller gains. These results indicate that slosh loads in the approach

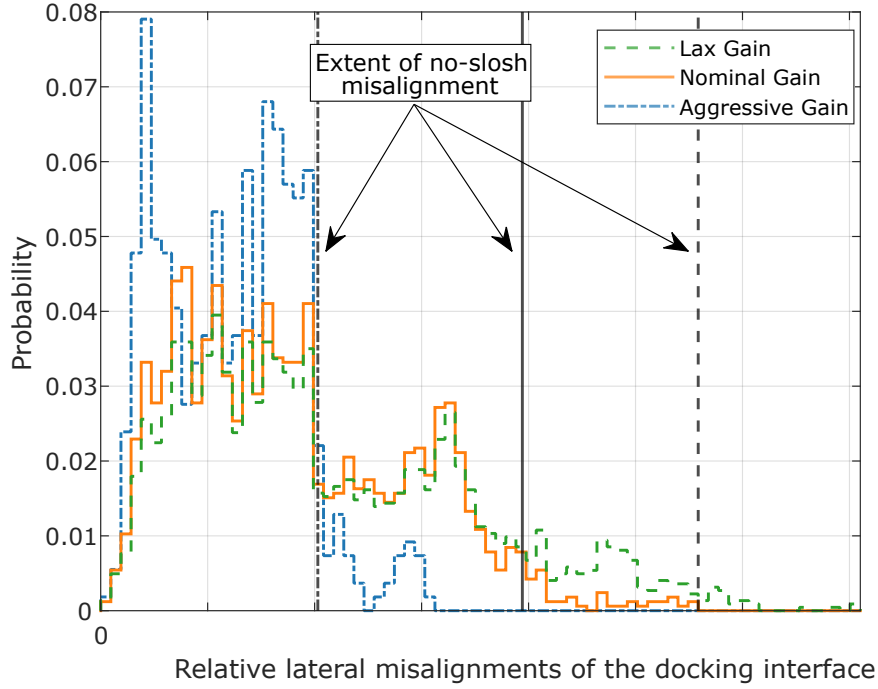


Figure 6: Relative lateral misalignment errors at the docking interface at the conclusion of the approach phase as a result of the Monte Carlo analysis.

phase have a significant influence on docking success due to their effect on misalignment ICCs.

Docking mechanism performance is analyzed using a multibody dynamics simulation of the NASA Docking System Block 2 (NDSB2), which is representative of an International Docking System Standard (IDSS)-compatible active docking mechanism. The NDSB2 model is coupled with the slosh particle model and the rigid body mass properties of the two vehicles, then initialized using the final vehicle and particle states from the approach phase simulation. For the integrated docking simulation, 100 of the worst-case ICCs were hand-selected from the approach simulation results for each of the slosh enabled and disabled datasets. These ICCs resulted in missed captures in the 17 out of the 100 cases with slosh disabled, and 19 out of 100 cases with slosh enabled. Additionally, 3 of the 100 slosh-enabled cases resulted in improper soft capture guide meshing and unintended contact between the soft capture system and the docking tunnel due to extreme vehicle misalignments at initial contact, whereas this did not occur in any of the slosh-disabled cases. ICCs and docking outcomes are shown in Figures 7 and 8 for the slosh-disabled and slosh-enabled simulations, respectively. Note that, as a result of hand-selecting the worst-case ICCs, the capture success rates are not indicative of the true likelihood of docking success — the true probability of capture success will likely be higher.

In the cases that captured successfully, propellant slosh forces persisted throughout soft captured operations through structural mating, as shown in Figure 9. These post-capture slosh loads had very low frequency relative to the structural modes of the coupled system and are within the capability of the soft capture system. These results suggest that the most significant consequences of propellant slosh are its effects on docking misalignments at initial contact, rather than on soft captured docking

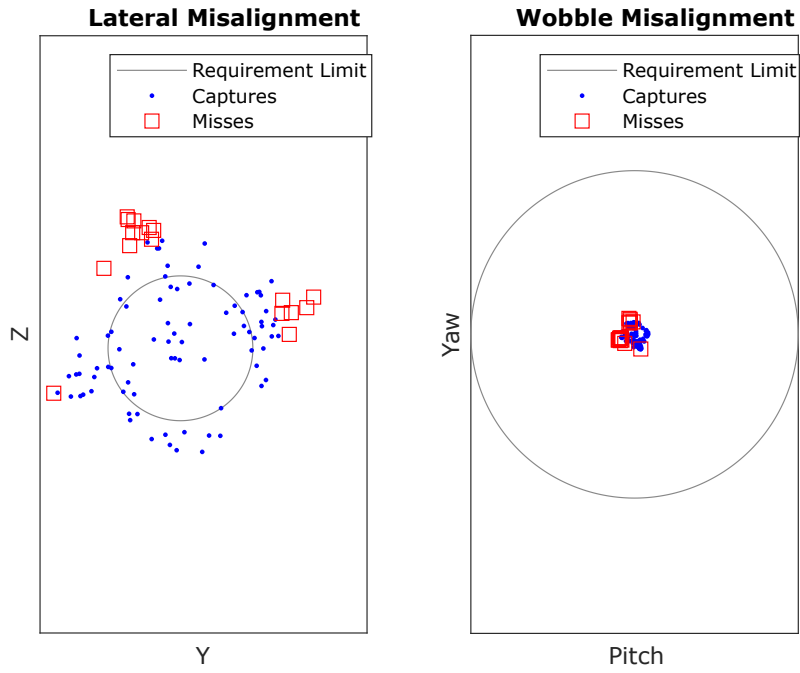


Figure 7: Soft capture results for cases with slosh disabled.

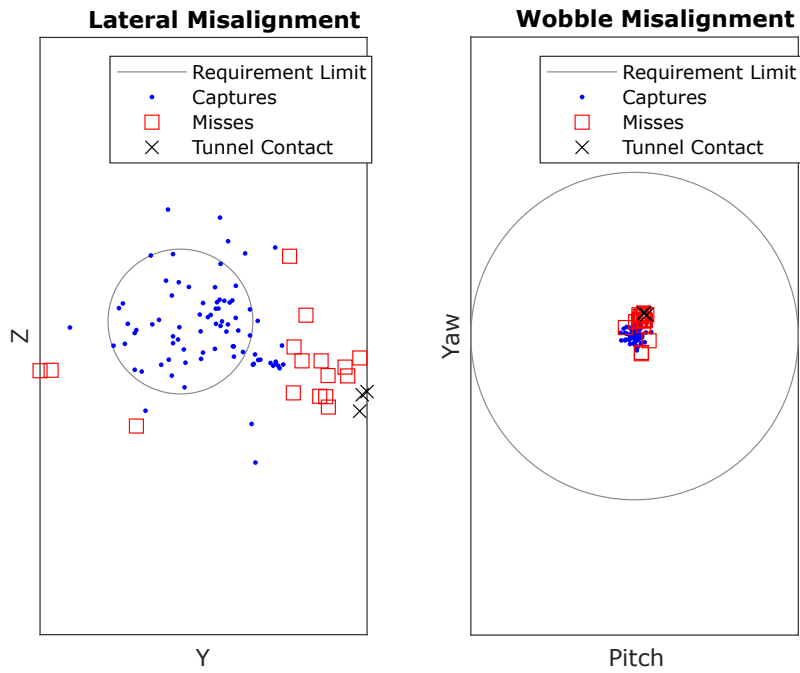


Figure 8: Soft capture results for cases with slosh enabled.

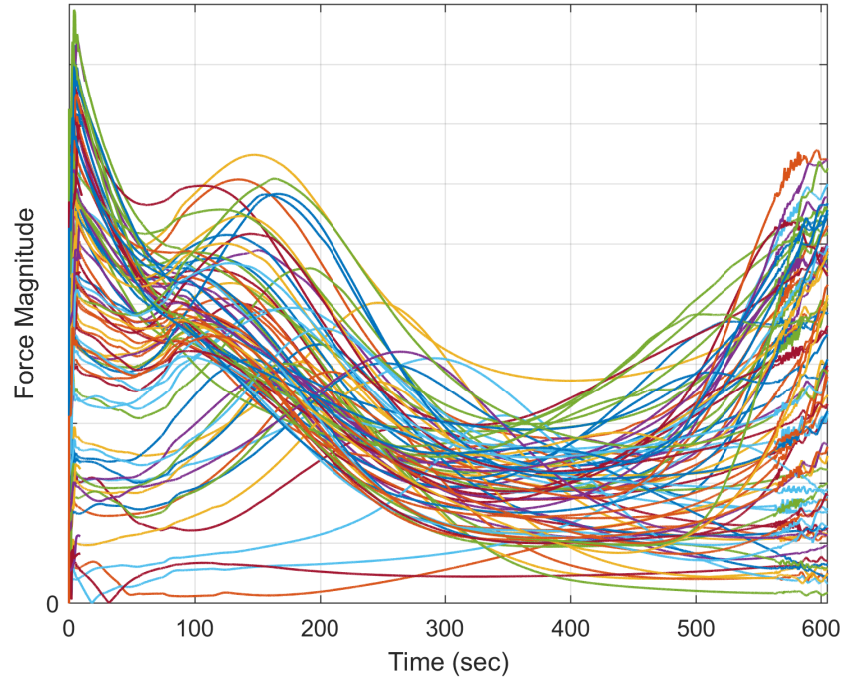


Figure 9: Total slosh force acting on chaser spacecraft during soft capture through structural mating.

operations.

CONCLUSION

A slosh model representing the liquid as moving and inert particles was presented. This work expands on the modeling approach for the single particle model described in previous literature through the introduction of 1) the inert particle that augments the spacecraft mass properties and 2) the smoothed contact dynamics. Two of the slosh model parameters (ω_n and EMF) were tuned using CFD simulation results. The slosh model was integrated with simulations of the approach and docking phases to assess the effects of slosh. During the approach phase, controller performance degraded with the inclusion of slosh-induced forces and torques. During the docking phase, slosh loads resulted in increased lateral misalignments at the docking interface at initial contact, which may lead to missed capture or damage to the docking mechanism in extreme cases. Slosh loads after capture were found to be within docking mechanism capability in this study. However, their inclusion in integrated docking verification is recommended due to their potential to influence mechanism alignment and structural mating.

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